

JEE MAIN REVISION MASTER: COMMUNICATION SYSTEMS OUT OF SYLLABUS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 98 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Amplitude Modulation & Sideband Analysis	72 PYQs	Modulation
Very High	EM Wave Propagation & Space Communication	26 PYQs	Wave Propagation

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Amplitude Modulation & Sideband Analysis	$\mu = \frac{A_m}{A_c} = \frac{A_{\text{max}} - A_{\text{min}}}{A_{\text{max}} + A_{\text{min}}}, \quad \text{Bandwidth} = 2f_m, \quad P_t = P_c \left(1 + \frac{\mu^2}{2}\right)$
EM Wave Propagation & Space Communication	$d_{\text{max}} = \sqrt{2Rh_T} + \sqrt{2Rh_R}, \quad f_c = 9\sqrt{N_{\text{max}}}, \quad f_{\text{MUF}} = f_c \sec\theta_i, \quad \text{Antenna length} \geq \frac{\lambda}{4}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Amplitude Modulation & Sideband Analysis (Very High Yield — 72 PYQs)

Core Principle: Principles of amplitude modulation, carrier and sideband components ($f_c \pm f_m$), modulation index, and power relationships.

■ **Caution / JEE Pitfall:** Modulation index μ must satisfy $\mu \leq 1$ (100%) to prevent envelope distortion.

■ **Speed Trick:** Carrier power does not depend on modulation index μ , only sideband power increases as μ increases.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "In an amplitude modulation, a modulating signal having amplitude of X Volt is superimposed with a carrier signal of amplitude Y Volt in firs..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "Match List I with List II : List I List II A. Attenuation I. Combination of a receiver and transmitter. B. Transducer II. process of retriev..." [Ans: N/A]
- [2023 Evening Shift] Q#2 (MCQ): "To radiate EM signal of wavelength λ with high efficiency, the antennas should have a minimum size equal to..." [Ans: N/A]

3.2 EM Wave Propagation & Space Communication (Very High Yield — 26 PYQs)

Core Principle: Propagation modes of electromagnetic waves (ground, sky, and space waves), critical frequency, MUF, and antenna LOS range calculations.

■ **Caution / JEE Pitfall:** Ensure earth radius $R \approx 6400\text{ km}$ and heights h_T, h_R are converted into the exact same units before square rooting.

■ **Speed Trick:** For an antenna of height h , maximum range is $d = \sqrt{2Rh}$, and population covered is $\pi d^2 \times \rho$.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "The height of transmitting antenna is 180m and the height of the receiving antenna is 245m . The maximum distance between them for satis..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "In satellite communication, the uplink frequency band used is :..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "The power radiated from a linear antenna of length l is proportional to (Given, $\lambda =$ Wavelength of wave):..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

The height of transmitting antenna is 180m and the height of the receiving antenna is 245m . The maximum distance between them for satisfactory communication in line of sight will be : (given $R=6400\text{ km}$)

- (A) 104km
- (B) 56km
- (C) 48km
- (D) 96km

Correct Answer: None

Solution Analysis: $d_{\text{(max)}} = \sqrt{2R h_T} + \sqrt{2R h_R}$ where $d_{\text{(max)}}$ is the maximum distance between the antennas, R is the radius of the earth, h_T is the height of the transmitting antenna above the earth's surface, and h_R is the height of the receiving antenna above the earth's surface. Substituting the given values, we get: $d_{\text{(max)}} = \sqrt{2 \times 6400\text{ km} \times 180\text{ m}} + \sqrt{2 \times 6400\text{ km} \times 245\text{ m}} = 104\text{ km}$ Therefore, the maximum distance between the transmitting and receiving antennas for line of sight communication is 104km .

Benchmark Problem 2: JEE Main 2023 (Online) 1st February Evening Shift (Q#1)

In an amplitude modulation, a modulating signal having amplitude of X Volt is superimposed with a carrier signal of amplitude Y Volt in first case. Then, in second case, the same modulating signal is superimposed with different carrier signal of amplitude $2Y$ Volt. The ratio of modulation index in the two cases respectively will be :

- (A) 2 : 1
- (B) 1 : 1
- (C) 4 : 1
- (D) 1 : 2

Correct Answer: None

Solution Analysis: The modulation index is a measure of the degree of modulation and is given by the ratio of the amplitude of the modulating signal to the amplitude of the carrier signal. In the first case, the modulation index is (X / Y) . In the second case, the modulation index is $(X / 2Y)$. The ratio of the modulation index in the two cases is $((X / Y) / (X / 2Y)) = (2 / 1)$